

2. Evaluation and Assessment of IR Systems

Outline

- 1 Motivation
- 2 Precision, Recall and MAP
- 3 Discounted Cumulative Gain
- 4 TREC and Further Evaluation Initiatives

Which system is better?

Google Information Retrieval Evaluation

About 87,900,000 results (0.52 seconds)

Scholarly articles for Information Retrieval Evaluation

Information retrieval evaluation - Hammar - Cited by 143
... significance tests for **information retrieval evaluation** - Smucker - Cited by 603
Novelty and diversity in **information retrieval evaluation** - Clarke - Cited by 1075

en.wikipedia.org - wiki - Evaluation_measures_...
Evaluation measures (information retrieval) - Wikipedia
Evaluation measures for an information retrieval system are used to assess how well the search results satisfied the user's query intent.
Offline metrics: Precision, Recall, F1 score, P-R curve, Precision at K

fig.standards.edu - IR book - IRIR - IRIR - IRIR - IRIR
Evaluation in information retrieval - Stanford NLP Group
Evaluation in information retrieval. We have seen in the preceding chapters many alternatives in designing an IR system. How do we know which of these ...

www.scribd.com - IR book - IRIR - IRIR - IRIR - IRIR
10. Evaluation in Information Retrieval - Universität Mannheim
Apr 20, 2020 - Learn how to evaluate information retrieval systems using the precision, recall and how it is leveraged in information retrieval evaluation.

www.dcc.ac.uk - Teaching - IRIR - IRIR - IRIR - IRIR
Lecture 6: Evaluation - Information Retrieval Computer ...
Today: how good are the retrieved documents? 227, Page 10, Overview, 1 Recap/Quiz, 2 Introduction, 3 Unsupervised evaluation, 4 Ranked evaluation, 5 ...

People also ask
How do you evaluate information retrieval?

Microsoft Bing Information Retrieval Evaluation

5,913,000 Results Date Language Region

Evaluation Metrics For Information Retrieval
<https://wwwnews.com/2020/05/information-retrieval-evaluation/>
Evaluation metrics for the Information Retrieval (IR) system are used to measure how well the search results satisfied the user's query intent. Such metrics are often split into two categories: online metrics (look at user's interaction with the search system, while offline metrics measure relevance, in other words how likely each result, or search engine results page (SERP) page as a whole, is to meet the information needs of the user.
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Evaluation measures (information retrieval) - Wikipedia
[https://en.wikipedia.org/wiki/Evaluation_measures_\(information_retrieval\)](https://en.wikipedia.org/wiki/Evaluation_measures_(information_retrieval))
Overview Notation Online metrics Offline metrics New work
Evaluation measures for an information retrieval system are used to assess how well the search results satisfied the user's query intent. Such metrics are often split into two categories: online metrics (look at user's interactions with the search system, while offline metrics measure relevance, in other words how likely each result, or search engine results page (SERP) page as a whole, is to meet the information needs of the user.
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Videos of Information Retrieval Evaluation
bing.com/videos

Information Retrieval Evaluation - 1.7k views - Jan 20, 2019
Information Retrieval Evaluation - 1.2k views - May 7, 2019
Information Retrieval Evaluation - 1.1k views - May 7, 2019

Evaluation measures
Information Retrieval
Evaluation measures for an information retrieval system are used to assess how well the search results satisfied the user's query intent. Such metrics are often split into two categories: online metrics (look at user's interaction with the search system, while offline metrics measure relevance, in other words how likely each result, or search engine results page (SERP) page as a whole, is to meet the information needs of the user.
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Support on edit

See results for
Information Retrieval
ICSE 2014, IR 2014, IRIR 2014, IRIR 2015, IRIR 2016, IRIR 2017, IRIR 2018, IRIR 2019, IRIR 2020, IRIR 2021, IRIR 2022, IRIR 2023, IRIR 2024, IRIR 2025, IRIR 2026, IRIR 2027, IRIR 2028, IRIR 2029, IRIR 2030, IRIR 2031, IRIR 2032, IRIR 2033, IRIR 2034, IRIR 2035, IRIR 2036, IRIR 2037, IRIR 2038, IRIR 2039, IRIR 2040, IRIR 2041, IRIR 2042, IRIR 2043, IRIR 2044, IRIR 2045, IRIR 2046, IRIR 2047, IRIR 2048, IRIR 2049, IRIR 2050, IRIR 2051, IRIR 2052, IRIR 2053, IRIR 2054, IRIR 2055, IRIR 2056, IRIR 2057, IRIR 2058, IRIR 2059, IRIR 2060, IRIR 2061, IRIR 2062, IRIR 2063, IRIR 2064, IRIR 2065, IRIR 2066, IRIR 2067, IRIR 2068, IRIR 2069, IRIR 2070, IRIR 2071, IRIR 2072, IRIR 2073, IRIR 2074, IRIR 2075, IRIR 2076, IRIR 2077, IRIR 2078, IRIR 2079, IRIR 2080, IRIR 2081, IRIR 2082, IRIR 2083, IRIR 2084, IRIR 2085, IRIR 2086, IRIR 2087, IRIR 2088, IRIR 2089, IRIR 2090, IRIR 2091, IRIR 2092, IRIR 2093, IRIR 2094, IRIR 2095, IRIR 2096, IRIR 2097, IRIR 2098, IRIR 2099, IRIR 2100, IRIR 2101, IRIR 2102, 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BMW R1100R - Wikipedia

The **BMW R 1100 R** is a standard motorcycle with a 1,000 cc (95.2 cu in) air-cooled flat twin engine, made by BMW Motorrad from 1994 through to 1999. Like all of the branch motorcycles of this period, all 53,685 units were made at BMW's Spandau, Berlin factory.

Production: 1994–1999 Manufacturer: BMW Motorrad

BMW engine: 95.0 mm × 70.5 mm (3.90 in × 2.78 in) Compression ratio: 10.5:1

[en.wikipedia.org · viki · BMW_R_1100R ·](#) + Translate this page

BMW R 1100 R - Wikipedia

Das **BMW R 1100 R** ist ein Motorrad des Fahrzeugherstellers BMW mit einem luft-gekühlten Zweizylinder-Motorblock. Das Modell ähnelt dem nächsten.

Leistung (DIN/PS): 57 kW (77.5 PS) bei 5.500 U/min Drehmoment (N·m): 37 Nm bei 3.250 min⁻¹

Abstufung (Stufen): 70–80 km/h (Stufenlos) Kraftstoffverbr.: 1,485 l/100 km

Konstruktion: Kasten Kategorie: Kasten

[en.wikipedia.org · viki · BMW_R_1100R ·](#) + Translate this page

BMW R1100R, Motorrad gebraucht und neu kaufen | eBay ..

eBay Kleinanzeigen: BMW R1100R, Motorrad gebraucht und neu kaufen oder verkaufen - Jetzt finden oder inserieren! eBay Kleinanzeigen - Kostenlos. Einfach. Lokal.

[www.ebay.de/kleinanzeigen.de · bmv ·](#) + Translate this page

Bmw R 1100 R, Motorrad gebraucht kaufen | eBay ..

eBay Kleinanzeigen: Bmw R 1100 R, Motorrad gebraucht kaufen oder verkaufen - Jetzt finden oder inserieren! eBay Kleinanzeigen - Kostenlos. Einfach. Lokal.

[www.1000ps.de · ... · BMW ·](#) + Translate this page

Gebrauchte und neue BMW R 1100 R Motorräder kaufen

Gebrauchte und neue **BMW R 1100 R** Motorräder auf 1000PS kaufen. Keine Provisionen!

 **BMW R1100R**

The BMW R 1100 R is a standard motorcycle with a 1,000 cc air-cooled flat twin engine, made by BMW Motorrad from 1994 through to 1999. Like all of the branch motorcycles of this period, all 53,685 units were made at BMW's Spandau, Berlin factory.

Fuel capacity: 21 l, 4.5 mpg gal (9.0 US gal)

Engine: 1,000 cc (95.2 cu in) air-cooled 4-stroke flat twin

Weight: 230 kg (506 lb), 340 kg (748 lb) (wet)

Top speed: 197 km/h (122 mph), 198 km/h (123 mph)

Power: 57 or 58 kW (76 or 79 hp), 56.7 kW (76.1 hp) @ 5,500 rpm

Torque: 39.4 N·m (28.9 ft·lb) @ 3,500 rpm

Brakes: Disk, 305 × 276 mm (12.0 × 10.9 in) front/rear, opt. ABS

Feedback

Google **"BMW R 1100 R"** 🔍 Sign in Phrase Search

[All](#) [Images](#) [Videos](#) [Shopping](#) [News](#) [More](#) [Settings](#) [Tools](#)

About 454,000 results (0.50 seconds)

Tip: Search for **English** results only. You can specify your search language in Preferences.

[en.wikipedia.org · viki · BMW_R_1100R ·](#) +

BMW R1100R - Wikipedia

The **BMW R 1100 R** is a standard motorcycle with a 1,000 cc (95.2 cu in) air-cooled flat twin engine, made by BMW Motorrad from 1994 through to 1999. Like all of the branch motorcycles of this period, all 53,685 units were made at BMW's Spandau, Berlin factory.

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[www.ebay.de/kleinanzeigen.de · bmv ·](#) + Translate this page

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[www.1000ps.de · ... · BMW ·](#) + Translate this page

Gebrauchte und neue BMW R 1100 R Motorräder kaufen

Gebrauchte und neue **BMW R 1100 R** Motorräder auf 1000PS kaufen. Keine Provisionen!

 **BMW R1100R**

The BMW R 1100 R is a standard motorcycle with a 1,000 cc air-cooled flat twin engine, made by BMW Motorrad from 1994 through to 1999. Like all of the branch motorcycles of this period, all 53,685 units were made at BMW's Spandau, Berlin factory.

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Power: 57 or 58 kW (76 or 79 hp), 56.7 kW (76.1 hp) @ 5,500 rpm

Torque: 39.4 N·m (28.9 ft·lb) @ 3,500 rpm

Brakes: Disk, 305 × 276 mm (12.0 × 10.9 in) front/rear, opt. ABS

Feedback

2.1 Motivation

- IR system evaluation goals:
 - ▶ **comparison** between several IR systems
 - ▶ IR system design and **configuration**
- Evaluation of database systems:
 - ▶ **functionality** (partly standardized: e.g. SQL)
 - ▶ runtime **efficiency** and storage space efficiency

Efficiency vs. Effectiveness

- **Efficiency** → using resources as sparingly as possible
 - ▶ typical in computer science: runtime and memory efficiency
 - ▶ “Is that what is being done being done correctly (efficiently)?”
 - ▶ comparative consideration of different systems only makes sense in case of equivalent or identical results
- **Effectiveness** → effectiveness means quality of a result in relation to the effort expended
 - ▶ “Are the right things being executed?”
- **Effectiveness** thus always aims at a technical solution in which **effort and benefit** are in a favorable relationship to each other.

Relevance

- When evaluating IR systems, one assumes that, with respect to a query q , the document collection falls into two parts:
 - ▶ R_q the set of documents relevant to the query
 - ▶ and $\overline{R_q}$ the set of documents not relevant to the query
- This assumption leads to the consequence that an IR system should provide exactly R_q .
- One can thus describe the quality of a query result by its deviation from R_q .

Problems of the binary definition of relevance

- The relevance depends on the **prior knowledge of the querier!**
 - ▶ The relevance of certain documents may only be recognized with prior knowledge!
 - ▶ On the other hand, if necessary, introductory overview articles are only interesting for the layman!
- The **relevance of documents can depend on each other.**
 - ▶ For example, the relevance of a document B may become evident only after reading document A.
 - ▶ Consequence: if A is in the result, B is relevant, otherwise not!
- A **binary relevance scale** seems to be too **simplistic.**
 - ▶ A multilevel or even a continuous scale would be more realistic.
 - ▶ But: It becomes even more difficult to find a generally accepted ranking of documents regarding a query!

2.2 Precision, Recall and MAP

	relevant	not relevant	Σ
in the result	a (hits)	b (noise)	a+b
not in the result	c (misses)	d (rejected)	c+d
Σ	a+c	b+d	a+b+c+d

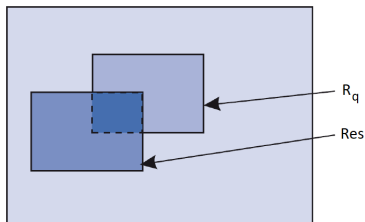
- Recall answers the question: How complete is the answer?

$$Recall = \frac{a}{a+c} = \frac{\text{Number of relevant documents in the result}}{\text{Total number of relevant documents}}$$

- Precision answers the question: How accurate is the answer?

$$Precision = \frac{a}{a+b} = \frac{\text{Number of relevant documents in the result}}{\text{Total number of documents in the result}}$$

Another representation



$$Recall = \frac{R_q \cap Res}{R_q}$$

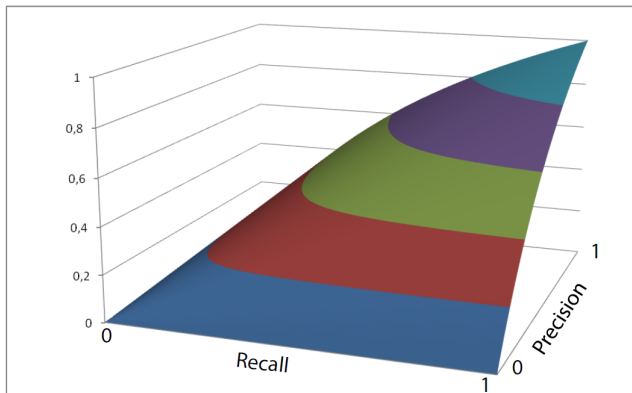
$$Precision = \frac{R_q \cap Res}{Res}$$

The F-Measure

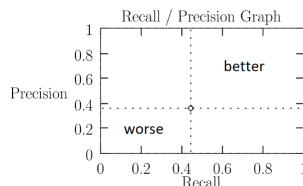
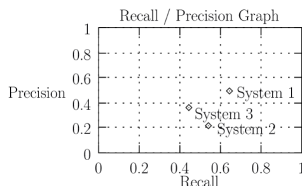
- **Combines** the two classic measures recall and precision
- Defines weighted **harmonic mean** of recall and precision
- Often the unweighted variant with $\alpha = 0.5$ is applied

$$F\text{-Measure} = \frac{Precision \cdot Recall}{(1-\alpha) \cdot Recall + \alpha \cdot Precision}$$

F-Measure



Comparison of multiple systems



Determination of Recall I

Variants:

- 1 Complete relevance assessment of a representative sample
 - ▶ problem: **proportion of relevant documents** is generally very low
 - ▶ $\Rightarrow$ the **sample** should be **very large**, which leads to a large effort

$$\text{Recall} = \frac{a}{a+c} = \frac{\text{Number of relevant documents in the result}}{\text{Total number of relevant documents}}$$

Determination of Recall II

2 Document Source Method

- ▶ choose a **random document** from the database
- ▶ **formulate a question** to which this document is relevant
- ▶ **check** whether the document is actually in the answer
- ▶ average over several documents and queries, the result is a value for the recall
- ▶ problem: questions are **not “real” user questions**

Determination of Recall III

③ Query extension

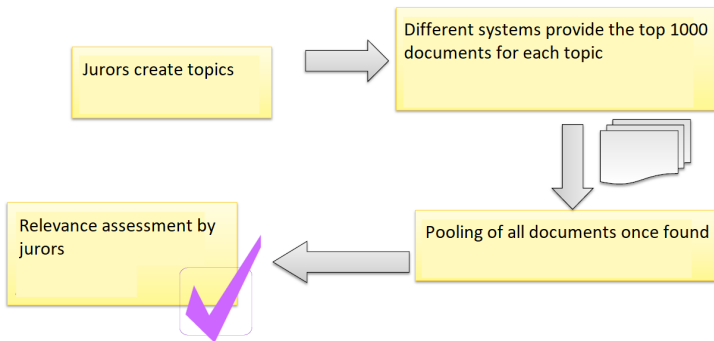
- ▶ expand the query so that a superset of the original answer set is found
- ▶ e.g. several **question formulations** from different editors
- ▶ problem: generally, you still only get a subset of the relevant documents!
- ▶ consequence: **recall estimate will generally be too high!**

Determination of Recall IV

4 Exploiting external sources

- ▶ try to determine relevant documents in parallel by using **independent methods**
- ▶ ask the questioner or other **professionals** what relevant documents they know

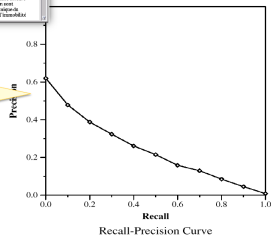
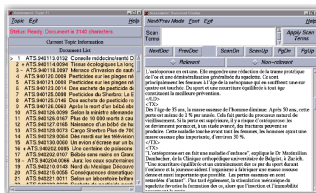
Pooling method



topics represent queries

Procedure

- Intellectual evaluation
 - ▶ relevant or not relevant
- Statistical evaluation





Source:TREC presentation slide No. 15

Text REtrieval Conference (TREC)

Distributed Information Retrieval Evaluation Campaign Tool (DIRECT) - Topic Selection - Mozilla Firefox

Datei Bearbeiten Ansicht Chronik Lesezeichen Extras Hilfe

http://direct.dei.unipd.it/protected/assessor/loadAssessorPool.do

Meistbesuchte Seiten Customize Links Free Hotmail Windows Media Windows

DIRECT

DISTRIBUTED INFORMATION RETRIEVAL EVALUATION CAMPAIGN TOOL

für das Cross Language Evaluation Forum (CLEF)

Angemeldet als mandl @ CLEF2008 [\[Abmelden\]](#)

Topic Selection

Abmeldung

Bewertungs-Hauptseite | User Management | Kontakte | Nachrichten

Bewertungs-Status	Name Datenbestand	Beschreibung Datenbestand	Bewertete Dokumente / Alle Dokumente
	10.2454/GC-ENGLISH-CLEF2008	GeoCLEF English Pool	14.528/14.528 100%
	10.2454/GC-GERMAN-CLEF2008	GeoCLEF German Pool	15.081/15.081 100%
	10.2454/GC-PORTUGUESE-CLEF2008	GeoCLEF Portuguese Pool	14.780/14.780 100%

★ Topic Kennung	📄 Topic Titel	🟢 Gesamt	🟢 Relevant	🔴 Nicht relevant	🟡 Nicht bewertet	Bewerten
10.2452/100-GC	Os documentos relevantes devem mencionar terramotos e enchentes nos estados da região oeste dos EUA.	744	31	713	0	Assess Topic
10.2452/76-GC	Documentos que mencionem a existência de um motim em prisões da América do Sul.	676	96	580	0	Assess Topic
10.2452/77-GC	Documentos que mencionem vencedores de Prêmios Nobel nascidos num país do Norte da Europa.	630	31	599	0	Assess Topic
10.2452/78-GC	Documentos que mencionem eventos desportivos que ocorram em (ou passem por) localidades, países ou territórios cobertos total ou parcialmente pelo deserto.	590	49	541	0	Assess Topic

Fertig

Distributed Information Retrieval Evaluation Campaign Tool (DIRECT) - Topic Assessment - Mozilla Firefox

http://direct.dsi.unipd.it/protected/assessor/assessment/assessTopic.do

Meistbesuchte Seiten Customize Links Free Hotmail Windows Media Windows

DIRECT Zurück

relevant Document << Document >> relevant Document >>
 nicht relevant noch nicht geprüft Document >> << nicht relevant Document <<

Topic

Information über Topic
 10.2452/100-GC:

Bewertete Dokumente / Alle Dokumente:
 715/715
 100%

Topic Titel
 Naturkatastrophen im Westen der USA

Topic Description
 Dokumente über Naturkatastrophen im Westen der USA

Topic Narrative
 Relevante Dokumente sollten Informationen über Erdbeben, Überschwemmungen und Waldbrände in den westlichen Bundesstaaten der USA enthalten. Zu den westlichen Bundesstaaten zählen Kalifornien, Washington und Oregon.

Nächstes Topic

Query Suche

Clean previous highlighted words ☐

Fertig

Dokumentkennung	Dokumenttitel	Relevanz
FR940109-000121	Nordkorea und USA Vor Druck gewarnt	nicht relevant

Dokumentkörper

<DOCNO>FR940109-000121</DOCNO>
 <WEEK> 940109 </WEEK>
 <TITLE> Nordkorea und USA </TITLE> <TITLE> Nordkorea und USA Vor Druck gewarnt </TITLE> <TEXT>
 WASHINGTON, 2. Januar (dpa). Ungeachtet einer Annäherung zwischen den USA und Nordkorea inderstrittenFrage der Atominspektionen hat der nordkoreanische Präsident Kim Il Sung Washington vor Druck und Drohungen gewarnt. Solch ein Vorgehen könnte zur Katastrophe führen, sagte Kim laut nordkoreanischer Nachrichtenagentur in einer Neujahresadresse in Pjöngjang. Nach dem jüngsten Gespräch zwischen Diplomaten beider Länder in New York hatte ein Sprecher des US-Außenministeriums trotz der offiziell geltenden Nachrichtensperre bestätigt, man sei dem Ziel nähergekommen. Wie der US-Sender ABC unter Berufung auf ungenannte Regierungsbeamte meldete, soll Nordkorea die Inspektion aller seiner insgesamt sieben Atomanlagen durch internationale Experten erlaubt und damit eine Hauptforderung der USA erfüllt haben. Nach nordkoreanischer Darstellung sollen die USA im Gegenzug bereit sein, auf die jährlichen Militärmanöver mit Südkorea zu verzichten. </TEXT>

Documents

FR940109-000121 - nicht relevant Nordkorea und USA Vor Druck gewarnt	View
Document 2 / 715 FR940109-000467 - nicht relevant 940109 SYDNEY, 7. Januar str. Die schwarze Buschfeuer in Australien haben am Freitag die Vororte de	View
Document 3 / 715 FR940109-001602 - nicht relevant senen zwischen Südkorea und in: Tokyo, Korea, die so auch Südkorea kommt	View

Documents

Averaging over multiple queries

Average over m experiments $\rightarrow$ 2 possibilities:

- 1. **Macro Assessment** (user-oriented approach)

$$Recall_{\phi,u} = \frac{1}{m} \cdot \sum_{i=1}^m \frac{a_i}{a_i + c_i} \quad Precision_{\phi,u} = \frac{1}{m} \cdot \sum_{i=1}^m \frac{a_i}{a_i + b_i}$$

- ▶ all experiments regardless of result size
- ▶ problem occurs when individual queries return empty results

$$a_i + b_i = 0$$

Averaging over multiple queries

- 2. **Micro Assessment** (system-oriented approach)
 - ▶ queries with a larger result set have more weight
 - ▶ sum of all queries considered as “one big query”

$$Recall_{\phi,s} = \frac{\sum_{i=1}^m a_i}{\sum_{i=1}^m (a_i + c_i)}$$

$$Precision_{\phi,s} = \frac{\sum_{i=1}^m a_i}{\sum_{i=1}^m (a_i + b_i)}$$

Example

Query	$ \text{Res}_q = a + b$	$ \text{R}_q = a + c$	$ \text{R}_q \cap \text{Res}_q = a$	Recall	Precision
1	10	7	1	0,14	0,1
2	12	13	2	0,15	0,17
3	200	138	120	0,87	0,6
4	4	9	2	0,22	0,5
Sum	226	167	125		

- Macro Assessment:

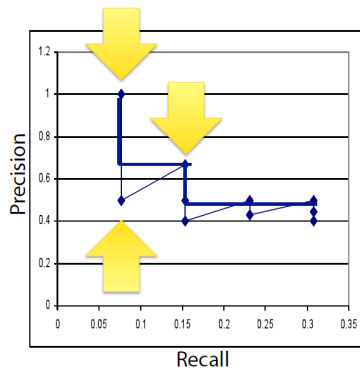
- ▶ $\text{Recall} : \frac{1}{4} \cdot (0.14 + 0.15 + 0.87 + 0.22) = 0.35$
- ▶ $\text{Precision} : \frac{1}{4} \cdot (0.1 + 0.17 + 0.6 + 0.5) = 0.34$

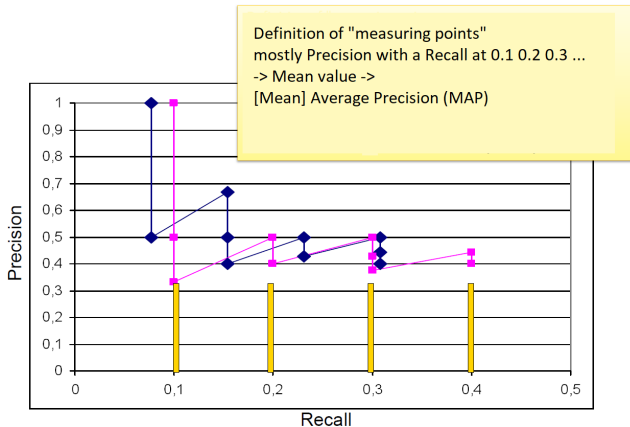
- Micro Assessment:

- ▶ $\text{Recall} : 125/167 = 0.75$
- ▶ $\text{Precision} : 125/226 = 0.55$
- “big” query 3 dominates

Recall/Precision for systems with ranking

Rank	Relevance
1	1
2	0
3	1
4	0
5	0
6	1
7	0
8	1
9	0
10	0





Important: user viewpoints and actually used metrics

- MAP → system-oriented approach
- But not directly interpretable as user viewpoint
- Check the documents only on the **first page**:
 - ▶ → P@10 or P@5
- Check the documents **until the first irrelevant one**:
 - ▶ good first hits are important
- Discounted Cumulative Gain
 - ▶ using the gradual relevance assessment by the user to measure the relevance gain caused by a document at a certain position

2.3 Discounted Cumulative Gain

- Basis: graded relevance from the user, e.g. $\{0, 1, 2, 3\}$ where 0 means irrelevant
- Assumption: highly relevant documents are most useful in terms of information gain when they reach a front position

$$DCG_p = rel_1 + \sum_{i=2}^p \frac{rel_i}{\log_2 i}$$

- rel_i is user relevance of i-th result document
- $\log_2 1 = 0$: avoid division by zero

Normalized DCG

- DCG_p of different runs only comparable after normalization → nDCG_p
- Idea: ratio to ideal DCG → IDCG_p
- IDCG_p: DCG_p sorted in descending order of user relevance

$$nDCG_p = \frac{DCG_p}{IDCG_p}$$

- Calculate average of nDCG_p from different experiments

Important: Query types

- Navigational
 - ▶ search for homepage of company X
- Informational
 - ▶ yellow pages requests
 - ▶ fact questions (question answering)
 - ▶ ad-hoc (search everything on topic X)
- Consequence: different evaluation requirements, especially for relevance feedback

Cranfield Experiment [CMK66]:

6 criteria for the evaluation of IR systems

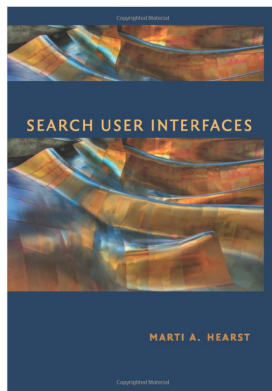
- ① **Recall** → ability of the system to find all relevant objects
- ② **Precision** → ability of the system to present only objects that are relevant
- ③ **Time lag** → the time between submitting the query and providing the response
- ④ **Effort** → the intellectual and physical effort required by the user to obtain the answer (e.g. complex input, queries from the system, ...)
- ⑤ **Form of presentation** → the quality of the result presentation, which influences the usefulness for the user
- ⑥ **Coverage of the collection** → the degree to which the system contains relevant objects (more a question of the documents than the retrieval system)

Vickery describes 6 criteria, which he divides into two groups [Vic70]:

- Criteria related to availability of information:
 - ▶ **Coverage** → proportion of potentially useful literature that was considered (e.g., if multiple collections were available: were all collections considered together with potentially relevant documents)
 - ▶ **Recall** → proportion of potentially useful documents included in the result of the query
 - ▶ **Response time** → average response time
- Criteria related to the selectivity of the response:
 - ▶ **Precision** → ability of the system to suppress useless documents
 - ▶ **Usability** → value of the determined documents for the user (reliability, comprehensibility, ...)
 - ▶ **Presentation** → form in which the results are presented

Usability

- Special focus in IR due to user focus
- <http://www.searchuserinterfaces.com/>



Goals of evaluation initiatives

- Uniform **evaluation standards** for retrieval systems (standardization)
- Enable **comparison** between systems
- Drive **further development** of IR systems
- Take up requirements from the community and improve methodology

2.4 TREC and Further Evaluation Initiatives

- TREC: Text REtrieval Conference
- “TREC is a new ballgame for IR research and development” (Sparck Jones 1994)
- Evaluation initiative of the National Institute of Standards and Technology (NIST) in the USA
- 1992: TREC-1 (Proceedings 1993)



Text REtrieval Conference (TREC)
*...to encourage research in information retrieval
 from large text collections.*



Overview

Other Evaluations

Publications

Information for Active Participants

Tracks

Past TREC Results

Contact Information

Frequently Asked Questions

Data

[Call for Participation in TREC 2021](#)
[Celebration of the 25th TREC: November 15, 2016](#)
[TREC Economic Impact Study](#)
[TREC Statement on Product Testing and Advertising](#)

The TREC Conference series is co-sponsored by the NIST [Information Technology Laboratory's \(ITL\) Retrieval Group](#) of the [Information Access Division \(IAD\)](#).
 Contact us at: [trec \(at\) nist.gov](mailto:trec@nist.gov)

NIST
 National Institute of Standards and Technology
 is an agency of the [U.S. Commerce Department](#)

Last updated: Wednesday, 16-Dec-2020 07:53:26 MST
 Date created: Tuesday, 01-Aug-00
[privacy policy / security notice / accessibility statement](#)
[disclaimer / FOIA](#)

Procedure

- Test base
 - ▶ objects (documents, ...) with clarified copyright
 - ▶ queries (Topics)
 - ★ relevant information needs for potential users
 - ▶ uniform relevance assessment
- Time frame (approx. 4 months in total)
 - ▶ query output
 - ▶ submitting results
 - ▶ announcement of the results

2008 TREC tracks

- Blog track
- Enterprise track
- Legal track
- Million query track
- Relevance feedback track

Further TREC tracks

- Cross-language track
- Filtering track
- Genomics track
- HARD track (high accuracy retrieval from documents)
- Interactive track
- Novelty track
- Question answering track
- Robust retrieval track
- SPAM track
- Terabyte track
- Video track
- Web track

Question answering

The screenshot shows the WolframAlpha interface. At the top, the logo 'WolframAlpha' is displayed with the tagline 'computational intelligence.'. Below the logo is a search bar containing the text 'scalar product'. To the right of the search bar are links for 'Extended Keyboard', 'Upload', 'Examples', and 'Random'. Below the search bar, a message states: 'Assuming "scalar product" refers to a computation | Use as referring to a mathematical definition or a word instead'. The 'Computational inputs' section shows two vectors: 'vector 1: {1, 2, 3}' and 'vector 2: {0, 4, 5}', each with a corresponding input field. A 'Compute' button is located below these inputs. The 'Input:' section shows the input '{1, 2, 3}, {0, 4, 5}'. The 'Result:' section shows the output '26'. A 'Step-by-step solution' button is available. Below the result, there is a 'Download Page' link and a 'POWERED BY THE WOLFRAM LANGUAGE' note. The 'Related Queries:' section lists several related mathematical queries, including 'sqrt(144), cos(0), sin(pi/2), ln(e^2), e^0', 'imager arithmetic', 'multiplication table', 'random animal curve', and 'Atanadikachimon-like curve vs Matthew Gray Gubi...'. At the bottom, there are links for 'Have a question about using Wolfram|Alpha?' and 'Give us your feedback'. The footer contains links for 'Pro', 'Web Apps', 'Mobile Apps', 'Products', 'Business Solutions', 'API & Developer Solutions', 'Resources & Tools', 'About', 'Contact', 'Connect', and 'English'.

Sample topic

<TOP>

<HEAD> Tipster Topic Description

<NUM> Number: 066

<DOM> Domain: Science and Technology

<TITLE> Natural Language Processing

<DESC>

Document will identify a ctype of natural language processing technology which is being developed or marketed in the U.S.

<NARR>

A relevant document will indentify a company or institution developing or marketing a natural language processing technology, identify the technology, and identify one or more features of the company's product.

<CON>NLP, translation, language, dictionary, font, software

<NAT> U.S.

</TOP>

TREC-3: without CON
TREC-4 and -5: only DESC
from TREC-5: variants
short and long

Sample document: Wall Street Journal

```
<DOC>
<DOCNO> WSJ880406-0090 </DOCNO>
<HL> AT&T Unveils Services to Upgrade Phone Networks Under
      Global Plan </HL>
<AUTHOR> Janet Guyon (WSJ Staff) </AUTHOR>
<DATELINE> New York </DATELINE>
<TEXT>
American Telephone & Telegraph Co. Introduced the first of a new
generation of phone services with broad ...
</TEXT>
</DOC>
```

Relevance evaluation

- Abstraction from the individual user and his context
- Uniform evaluation
- Objective jurors who are not in a use situation
- Objective statement about content relation between topic and document

Cross-language evaluation forum

Research on cross- and multi-lingual
Information Retrieval systems



EU funding: DELOS NoE for digital
libraries

Test environment

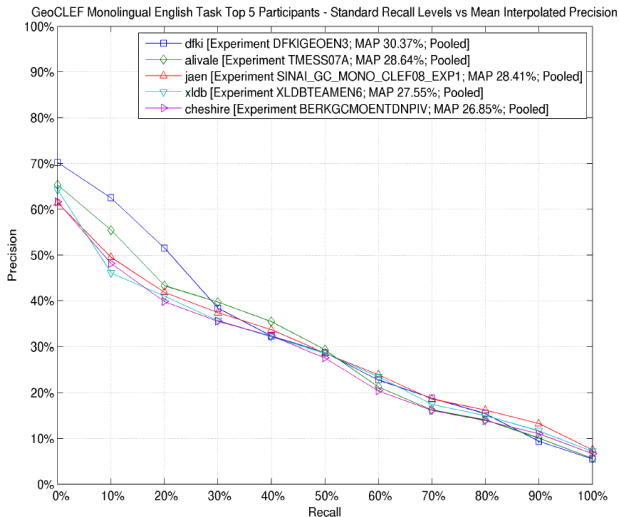
System development

Benchmarks

Evaluation research

Mandl et al. @ CLEF 2003 - 2006

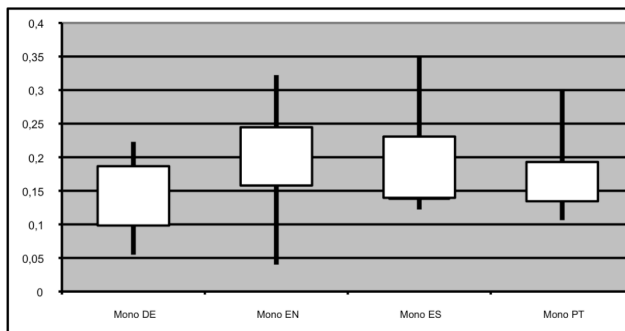
GeoCLEF Monolingual English



Variability

- **OBSERVATION:** the difference in performance between the systems is much smaller than between the topics

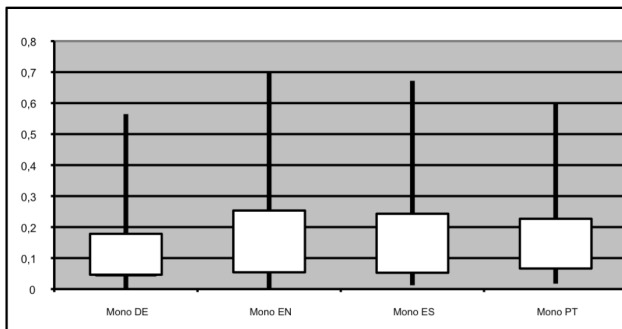
Example GeoCLEF 2006 (boxplots) I



Variance of the **systems** in GeoCLEF 2006

Mandl 2008

Example GeoCLEF 2006 (boxplots) II



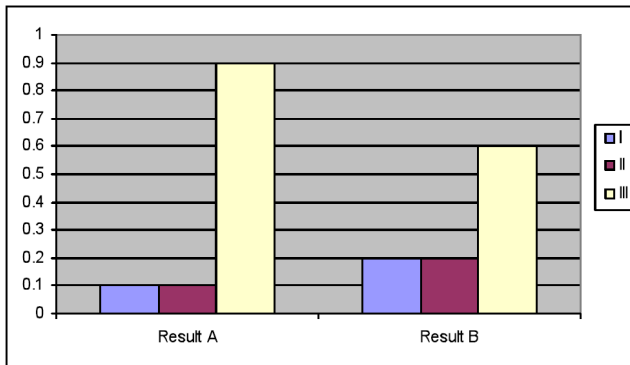
Variance of the **topics** in GeoCLEF 2006

Mandl 2008

Aggregation of multiple queries

- Usually simple arithmetic average

Which system is better?



2 systems, 3 queries

“IR Psychology” I

The **unhappy customer**, on average, **will tell 27** other people ...

Dissatisfied customers tell an average of **ten** other people about their bad experience. Twelve percent tell up to twenty people.

→ **Bad news travels fast.**

Credit to Jacques Savoy for slide

“IR Psychology” II

On the other hand, satisfied customers will tell an average of *five* people about their positive experience.

→ **Good news travels somewhat slower.**

A site search should work robustly

~> bad outliers must be avoided!

for each query

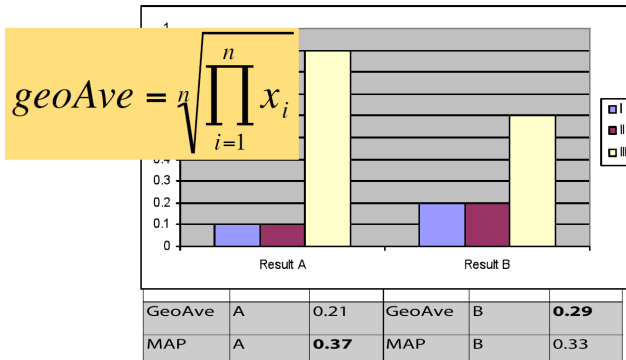
for each evaluation criterion

Credit to Jacques Savoy for slide

“IR Psychology” III

- Very good results, poor interaction → dissatisfied user
- Very good interaction, poor results → dissatisfied user

GMAP



Robustness

- Stable performance over all queries instead of high average performance
- “Difficult” queries should have greater weight in the evaluation process

NTCIR

- Multilingual retrieval for Asian languages
- Performed in Tokyo
 - ▶ National Institute for Informatics
- Tasks
 - ▶ cross-lingual
 - ★ Chinese, Japanese, Korean → English
 - ▶ patent retrieval
 - ▶ web retrieval
 - ▶ question answering

FIRE

- Evaluation Initiative for Indian Languages
 - ▶ Hindi and Bengali (among the 10 most spoken languages in the world)
 - ▶ Marathi, Tamil, Telugu, Punjabi, Malayalam
 - ▶ English
- Newspaper texts
- Running since 2008
- <https://www.isical.ac.in/~clia/>

INEX

- Initiative for the Evaluation of XML Retrieval

- ▶ evaluation of retrieval with structured documents
- ▶ goal: find the smallest relevant section rather than the entire document
- ▶ exploiting the structure
- ▶ test collection 2005: travel guide

- ★ <https://www.is.inf.uni-due.de/projects/inex/index.html>

IMIRSEL

- International Music Information Retrieval Systems Evaluation Laboratory Project
 - ▶ answering questions about pieces of music
- https://www.music-ir.org/mirex/wiki/MIREX_HOME

User?

- The user is never in the role of the evaluator
- Does improving systems in terms of traditional metrics also lead to higher user satisfaction?
- Does a good ranking also lead to higher user satisfaction?

Problems with user satisfaction I

- No correlation between preference and quality could be detected in several image retrieval systems
 - ▶ two tasks
 - ▶ eleven test subjects

Maskari et al. 2006

Problems with user satisfaction II

- For simple search tasks and controlled retrieval quality, there was no effect on user performance for a system
 - ▶ artificially created systems of different quality (different MAP)
 - ▶ as many relevant documents as possible should be found in 5 minutes
 - ▶ approx. 20 test subjects

Turpin & Scholer 2006

- System differences cannot be identified by user testing
 - ▶ model for TREC and CLEF is the “dedicated searcher”

Turpin & Hersh 2001